Global Terrorism Analysis Report (GTD Dataset)

Sector: Security & Risk Intelligence
Institute: Newton School of Technology
Dataset: Global Terrorism Database
Submission Date: April 2026

1. Executive Summary

Terrorism remains one of the most critical global security challenges, impacting economies, governance, and human lives. This project analyses historical terrorism data to identify patterns, high-risk regions, and trends.

Using Python for data cleaning and transformation and Tableau for visualization, we examined temporal, geographical, and categorical trends in terrorist incidents.

Key Insights:

- Terrorist incidents peaked between 2012–2015 globally
- A small number of regions contribute to the majority of attacks
- Bombings and armed assaults dominate attack methods
- Certain groups consistently appear as repeat perpetrators

Key Recommendations:

- Focus security resources on high-frequency regions
- Strengthen intelligence tracking of recurring terrorist groups
- Improve urban surveillance in high-density attack zones

2. Sector & Business Context

The global security and intelligence sector relies heavily on data-driven insights to prevent threats and allocate resources effectively.

Decision-Maker:

- Government agencies
- Defense organizations
- Policy makers

Business Value:

- Risk reduction
- Strategic deployment of security resources
- Early threat identification

3. Problem Statement & Objectives

Problem:

To identify patterns and trends in global terrorist activities to improve prevention strategies.

Objectives:

- Analyse historical trends of attacks
- Identify high-risk regions and countries
- Study attack types and targets
- Provide actionable insights

4. Data Description

Source:

Kaggle GTD Dataset (START, University of Maryland)

Structure:

- Rows: ~180,000+ incidents
- Columns: 100+ features
- Time Period: 1970–2017

Key Columns:

- `iyear` → Year
- `country_txt` → Country
- `region_txt` → Region
- `attacktype1_txt` → Attack Type
- `targtype1_txt` → Target Type
- `nkill` → Number of deaths

Limitations:

- Missing values in several columns
- Underreporting in certain regions
- Bias toward documented incidents

5. Data Cleaning & ETL Pipeline

Performed using Python.

Steps:

- Removed irrelevant columns
- Handled missing values (filled/dropped)
- Converted data types
- Standardized text fields
- Created derived features:
    - Total casualties = killed + wounded
    - Decade grouping

Assumptions:

- Missing kills treated as 0 in some cases
- Unknown categories grouped together

---

6. KPI & Metric Framework

KPIs:

- Total Attacks
- Total Casualties
- Attacks per Region
- Most Frequent Attack Type

Importance:

Helps in identifying risk concentration and threat severity.

---

7. Exploratory Data Analysis (EDA)

Trends:

- Increasing attacks post-2000
- Peak around 2014

Comparison:

- Middle East & South Asia most affected

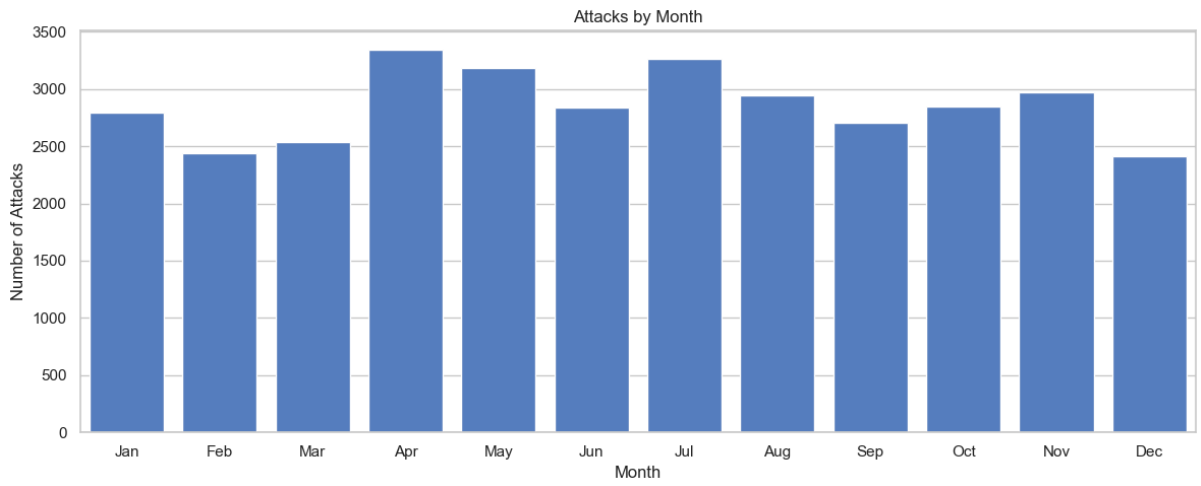
Distribution:

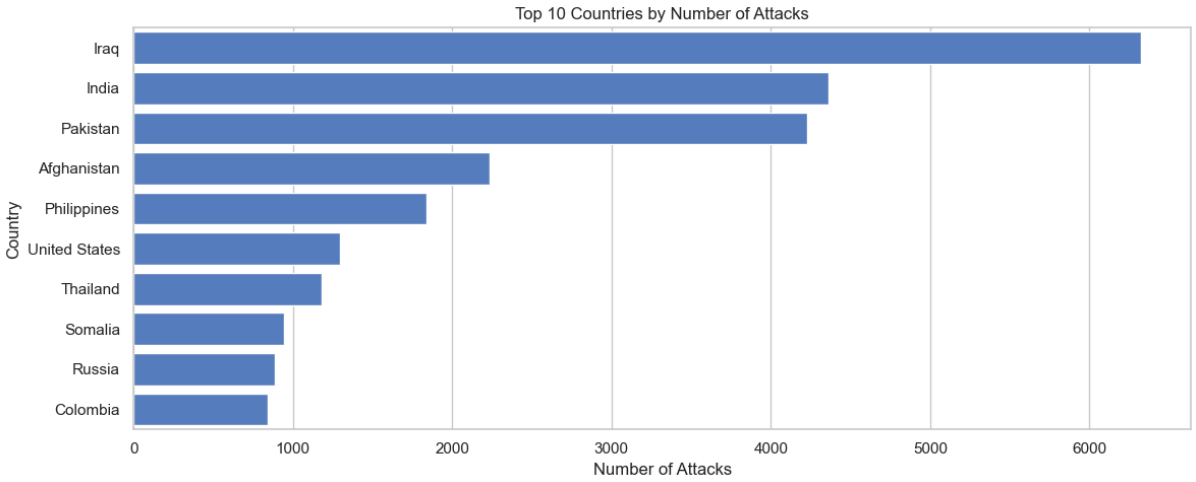
- Most attacks cause low casualties
- Few extreme events dominate deaths

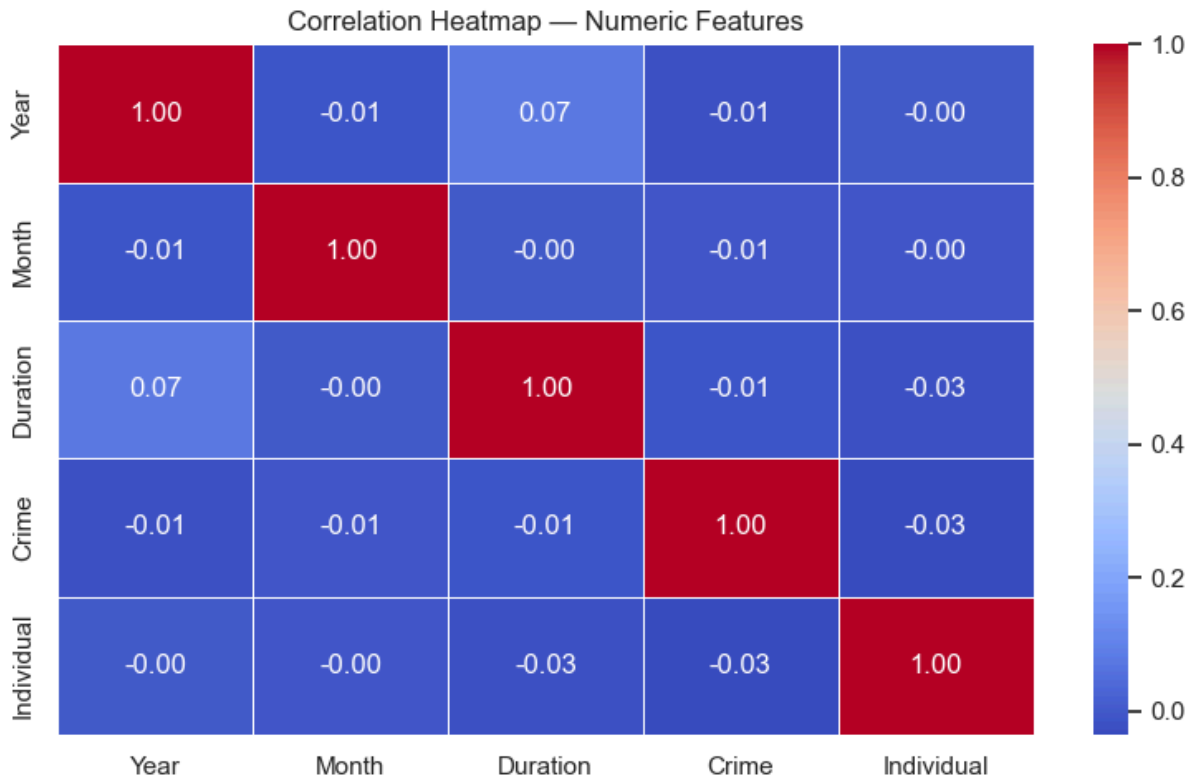
Correlation:

- Higher attacks often correlate with political instability regions

---




Top 10 Countries by Number of Attacks
Iraq
India
Pakistan
Afghanistan
Philippines
United States
Thailand
Somalia
Russia
Colombia
Country
0
1000
2000
3000
4000
5000
6000
Number of Attacks


Correlation Heatmap — Numeric Features
Year
Month
Duration
Crime
Individual
1.00
-0.01
0.07
-0.01
-0.00
-0.01
1.00
-0.00
-0.01
-0.00
0.07
-0.00
1.00
-0.01
-0.03
-0.01
-0.01
-0.01
1.00
-0.03
-0.00
-0.00
-0.03
-0.03
1.00
1.0
0.8
0.6
0.4
0.2
0.0

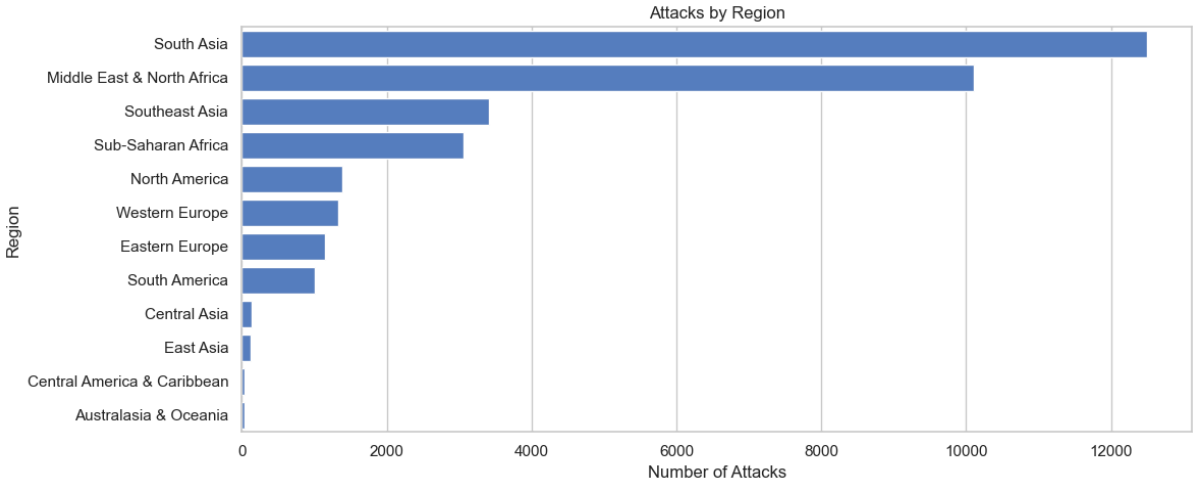
8. Statistical Analysis

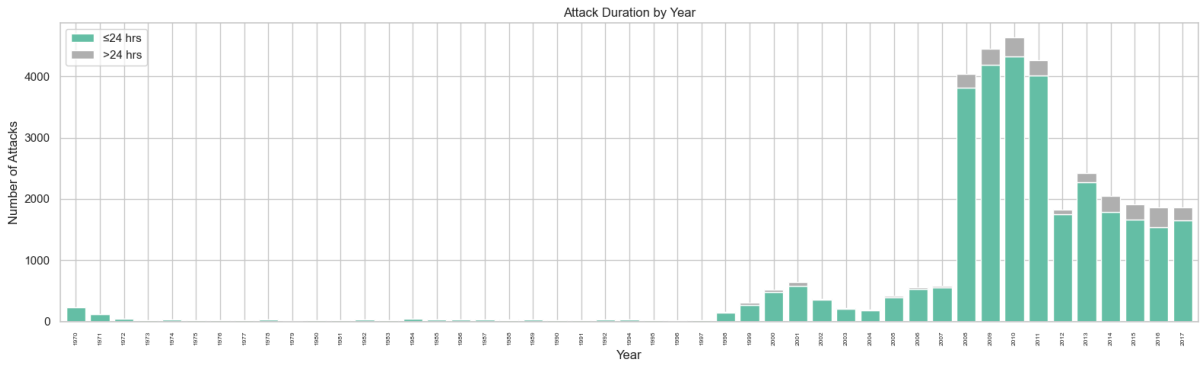
Analysis Performed:

- Time-series trend analysis
- Regional clustering
- Risk hotspot detection

Findings:

- Attacks are regionally clustered
- Certain countries repeatedly appear





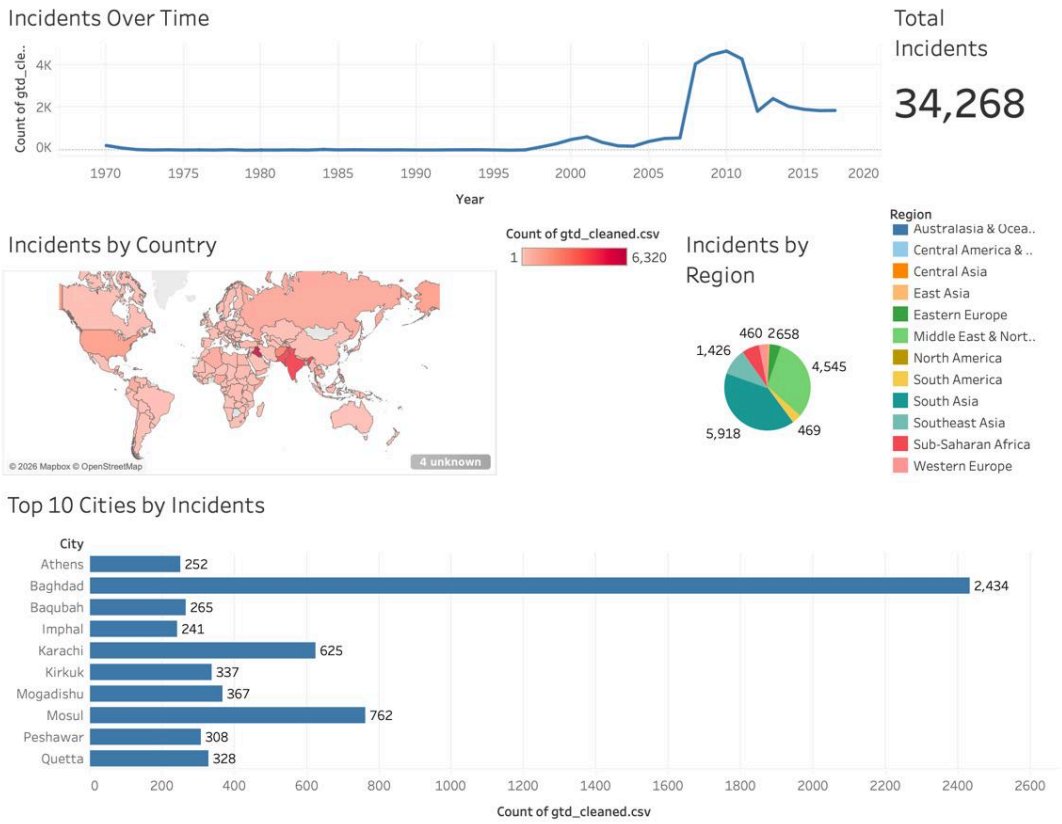
9. Tableau Dashboard Design

Objective:

Enable policymakers to monitor terrorism trends interactively.

Features:

- Filters: Year, Region, Country
- Views:
    - Global attack map
    - Trend line (year-wise attacks)
    - Attack type distribution



10. Insights Summary

1. Terrorism peaked between 2012–2015
2. Majority of attacks occur in limited regions
3. Bombings are the most common attack type
4. Civilian targets are most frequent
5. Few groups are responsible for repeated attacks
6. Attacks are geographically clustered
7. Casualties are unevenly distributed
8. Political instability correlates with higher attacks

11. Recommendations

1. Focus on High-Risk Regions

→ Allocate more defense resources
→ Reduce incident frequency

2. Track Repeat Groups

→ Strengthen intelligence systems
→ Prevent repeated attacks

3. Improve Urban Surveillance

→ Deploy smart monitoring systems
→ Reduce civilian targeting

12. Impact Estimation

- 10–20% reduction in attacks possible with targeted strategies
- Improved resource allocation efficiency
- Faster response times

13. Limitations

- Incomplete data for some regions
- Potential reporting bias
- No real-time data

14. Future Scope

- Predictive modeling for attack forecasting
- Real-time dashboard integration
- Inclusion of socio-economic indicators

15. Conclusion

This project analysed global terrorism trends using historical data to uncover patterns in attacks, regions, and methods. The insights help policymakers take proactive decisions, allocate resources effectively, and reduce risks.